Check of your `Evolution_two_rho.tex`, `Evolution_three_rho.tex` and `Evolution_terms.tex`

Notation as in your files: $\mathcal{K}(X) = X/X^2$, $U^{bc} = U^{cb}$. Blue: new or corrected; red: removed. Reference: the $\log \frac{k^+}{\Lambda}$ part (the "J" list) of every row of your current `Group_I.tex`, `Group_II.tex`, `Group_III.tex`, obtained by the exact split of `Checks_JIMWLK_BFKL.pdf` (each row $= \log \frac{k^+}{\Lambda}\, J + \log \frac{\vee}{k^+}\, B + R$).

Method. Every equation of your three files is evaluated pointwise: random transverse points, random classical charges, $U = U^T$, $\log \frac{k^+}{\Lambda} = 1$, and $\epsilon = 0.37$ kept finite in the pole brackets (so $d_\perp = 2 - \epsilon$ is tested too). A three-ρ term is summed over the 6 placements of its three charges. Groups I and III are taken with their c.c. (conjugate with $w \leftrightarrow w'$), and Group II without. Two expressions agree when the relative difference is at the level of rounding, 10^{-13} or below.

1 Answer

- `Evolution_two_rho.tex` **is correct.** All seven displayed rows equal the two-ρ $\log \frac{k^+}{\Lambda}$ terms of your Group rows exactly (Table 1). This includes the new Group I Row V 2b split (UV and finite), the new $d_\perp$ bracket of Group II Row IV, and the sign $-$ of Group III Row III part 2. Only two measures need fixing (T1, T2).
- `Evolution_three_rho.tex`**:**
  - Every row is correct except Group I Row III, d^3N_2 (E1, E2; Table 2). Group I Row II now has fix F4 and agrees exactly.
  - The list "All three ρ term with c.c." agrees row by row with the Group rows, *including* $\mathcal{D}$ in Group I Row III, and cancels as a whole: 1.3×10^{-13} against terms of size 120–260 (Table 3).
  - All twelve $\alpha, \alpha', \beta, \beta'$ blocks of "Cancellation of all three ρ's" vanish ($\leq 1.2 \times 10^{-13}$).
  - The rest are typos (T3–T8).
- `Evolution_terms.tex` **is correct.**
  - It is the previously checked file with $\log \frac{\vee}{\Lambda} \to \log \frac{k^+}{\Lambda}$ and a new title; nothing else changed.
  - Sec. 1 + Sec. 2 = Sec. 3 = JIMWLK$\otimes$LO to 8×10^{-17} (Sec. 4).
  - Two typos (T9, T10) and one remark on the commented-out BFKL section.

Corrected copies are in `corrected/`: `*_marked.pdf` shows the changes (red/blue), and `*.pdf` is clean. The clean copies were re-checked with the same programs: every row agrees ($\leq 2.4 \times 10^{-13}$), and the list still cancels.

2 `Evolution_two_rho.tex`

your row	your Group row (J part)	relative difference
Group I, Row II	G1 II part 1	0
Group I, Row V, d^3N_{2a}	G1 V 2a	2.0×10^{-16}
Group I, Row V, $d^3N_{2b\,\mathrm{UV}}$	G1 V 2b UV	0
Group I, Row V, $d^3N_{2b\,\mathrm{finite}}$	G1 V 2b finite	1.6×10^{-16}
Group II, Row I	G2 I part 2	0
Group II, Row IV	G2 IV	0
Group III, Row III, second part	G3 III part 2	0

Table 1: Every two-ρ row against the $\log \frac{k^+}{\Lambda}$ two-ρ terms of your Group rows (three seeds; poles at $\epsilon = 0.37$).

T1 Group I, Row II: $\int_{w,w',x,x',}$ (stray comma).

T2 Group III, Row III, second part: $\int_{w,y',z,y,x} \to \int_{w,x',y',z,x}$. There is no y in the integrand, and x' is integrated.

3 Evolution_three_rho.tex

3.1 Row by row

your row	form	as written	corrected
Group I, Row II	(z form and log form)	2.2×10^{-16}	2.2×10^{-16}
Group I, Row III	first form ($d^3N_1 + d^3N_2$)	differs (E1)	6.9×10^{-14}
Group I, Row III	rewritten form	differs (E2)	3.1×10^{-14}
Group I, Row IV	first / rewritten	1.2×10^{-15} / 1.4×10^{-15}	same
Group I, Row V	first / rewritten	2.4×10^{-16} / 4.9×10^{-16}	same
Group II, Row I	first / rewritten	2.8×10^{-13} / not readable (T5)	2.8×10^{-13} / 2.4×10^{-13}
Group III, Row I	first / rewritten	7.5×10^{-16} / 1.1×10^{-15}	same
Group III, Row II	first / rewritten	8.0×10^{-16} / 5.1×10^{-16}	same
Group III, Row III, first part	first / rewritten	0 / 3.8×10^{-16}	same
Group III, Row V	first / rewritten	0 / 1.0×10^{-14}	same
Group III, Row VI	first / rewritten	0 / 9.5×10^{-15}	same

Table 2: Every three-ρ row (symmetrized over the charges; with c.c. for Groups I, III) against the log $\frac{k^+}{\Lambda}$ three-ρ terms of your Group rows. Group I Row III is compared with $\mathcal{D}$ (fix F3).

3.2 Group I, Row III (E1, E2)

Write $\mathcal{P} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\,\frac{ig^4 f^{abd}}{\pi^5}\,\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}$, and let $A = \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]$, $B_z = \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$ be the two colour brackets of your file. The kernels are

$$Q_1 = \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2}, \qquad Q_2 = \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2},$$

$$E = \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\ \text{(edge kernel)}, \qquad P = \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(x-z)^2+(x-w)^2}{[(x-z)^2-(x-w)^2]^2}.$$

The log $\frac{k^+}{\Lambda}$ terms of Row III in your `Group_I.tex` are, block by block:

$$d^3N_1:\ \mathcal{P}\Big[\tfrac{1}{16}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z) - \tfrac{1}{16}P + \tfrac{1}{16}P\Big]A\,B_z, \tag{3.1}$$

$$d^3N_2:\ \mathcal{P}\Big(-\tfrac{1}{8}\Big)[Q_1-Q_2]\,A\,B_z \qquad \text{(the bracket has the prefactor } -\tfrac{1}{(2\pi)^3}\text{ in your file)}, \tag{3.2}$$

$$d^3N_{3a}:\ \mathcal{P}\,\tfrac{1}{8}[-E-Q_2+Q_1]\,A\,B_z. \tag{3.3}$$

The P terms cancel ($+\log-\log$), and Q_1, Q_2 of d^3N_2 cancel those of d^3N_{3a} term by term. What is left is

$$\text{Row III}\big|_{\log\frac{k^+}{\Lambda}} = \mathcal{P}\Big[\tfrac{1}{16}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,A\,B_z - \tfrac{1}{8}\,E\,A\,B_z\Big], \tag{3.4}$$

and with fix F3 the edge term becomes $\mathcal{D}$:

$$-\tfrac{1}{8}\,E\,A\,B_z \ \to\ -\tfrac{1}{8}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,A\,B_w. \tag{3.5}$$

This is exactly what your list "All three ρ term with c.c." uses (Table 3).

E1 First form, d^3N_2. You write $+\frac{1}{8}[Q_1-Q_2]$, but it is $-\frac{1}{8}[Q_1-Q_2]$. As written, the five terms add up to $\frac{1}{8}[2Q_1-2Q_2-E]$ instead of $-\frac{1}{8}E$. Corrected:

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ &-\frac{ig^4f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'} Q_1\,A\,B_z \\ &+\frac{ig^4f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'} Q_2\,A\,B_z \\ &-\frac{ig^4f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\,\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2(y-w)^2(x-z)^2(w-z)^2}\,A\,B_{zw} \\ &-\frac{ig^4f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'} Q_2\,A\,B_z + \frac{ig^4f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'} Q_1\,A\,B_z\Bigg\}.\end{aligned} \tag{3.6}$$

The Q_1 and Q_2 terms now cancel in pairs, and the remaining (third) line is $\mathcal{D}$.

E2 Rewritten form, d^3N_2. As written it is $-\frac{1}{16}\,\mathcal{P}\,P\,A\,B_z$, the P term that cancels inside d^3N_1. After symmetrization it is exactly half of your first form as written. Corrected:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4f^{abd}}{168\pi^5}\,\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,w,w',x'} e^{-ik(w'-w)}$$

$$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-w)^2-(x-z)^2\right]^2}$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(zw)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]. \tag{3.7}$$

With E1 and E2, both forms of Row III equal the Group I row with $\mathcal{D}$ (6.9×10^{-14} and 3.1×10^{-14}). Your d^3N_1 (both forms) is correct.

3.3 Typos

T3 Group I, Row II, second (log) term: there is no z left, so $\int_{x,y,x',z}$.

T4 Group I, Row V, first form: $\int_{w,z,y,x',y',w'}$.

T5 Group II, Row I, rewritten form: four terms read $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$. A product of four vectors joined by three dots is not defined. It should be $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$, and then the row is correct (2.4×10^{-13}).

T6 "From four ρ's", d^3N_a and d^3N_b:

- This file is the log $\frac{k^+}{\Lambda}$ (JIMWLK) part, so the factor is $\big[\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}\big]\to\log\frac{k^+}{\Lambda}$.
- The log $\frac{\vee}{k^+}$ part of these terms belongs to the BFKL list, where it cancels too (`Checks_JIMWLK_BFKL.pdf`).
- Your cancellation section already uses log $\frac{k^+}{\Lambda}$ only.

T7 Genuine terms, Group I Row II, d^3N_2: the closing `\Bigg]` is missing.

T8 Genuine terms, Group III Row III: two terms begin with `\-`, a discretionary hyphen that prints nothing. The minus sign is missing in the PDF; write `-`.

3.4 The list "All three ρ term with c.c." and the cancellation section

part of the list	against (relative difference)
From four ρ's ($d^3N_a+d^3N_b$)	three-ρ lists of `4rho.tex`: 3×10^{-16}
Genuine, Group I Row II / III / IV / V	Group rows + c.c.: 2×10^{-16} / 3×10^{-14} ($\mathcal{D}$) / 3×10^{-15} / 2×10^{-15}
Genuine, Group II Row I	Group row: 2.4×10^{-13}
Genuine, Group III Row I / II / III / V / VI	Group rows + c.c.: 3×10^{-14} / 2×10^{-15} / 4×10^{-15} / 3×10^{-13} / 9×10^{-14}
whole list (symmetrized)	1.3×10^{-13} (largest term 120–260), in units of $\frac{g^4}{(2\pi)^3 16\pi^5}$
each of the 12 blocks $\alpha,\alpha',\beta,\beta'$ (two, three, four Wilson lines)	$\le 1.2\times10^{-13}$ (direct and with c.c.)

Table 3: The list is written with its c.c., so each row is compared with the Group row plus its c.c. The list is the same as before (it was shown to cancel then too), and now with log $\frac{k^+}{\Lambda}$.

4 `Evolution_terms.tex`

Apart from the title and log $\frac{\vee}{\Lambda}\to\log\frac{k^+}{\Lambda}$ (32 lines), the file is identical to the one checked before. The check was re-run on the new file:

Sec. 1 (8 equations, 139 terms) + Sec. 2 (16 equations, 365 terms) vs Sec. 3 ($d^3N_1+d^3N_2$)	6.7×10^{-17}
Sec. 1 + Sec. 2 vs Sec. 3 (collected form)	8.4×10^{-17}
Sec. 3 (collected form) vs JIMWLK⊗LO	1.8×10^{-16}
the two-ρ "From 4ρ's" list vs the two-ρ lists of `4rho.tex` (with F1, F2)	1.2×10^{-16}
left-over |Sec. 1+Sec. 2−Sec. 3| / $\sum$|terms|	8.1×10^{-17}

The colour-singlet forms are unchanged from the version checked before (correct for any U). Their prefactor now carries $\frac{1}{N_c^2-1}\langle\rho^f(x)\rho^f(y)\rangle$.

T9 Colour-singlet form (and the same line in the commented-out BFKL section): $\int_{x,y,z,w,w'}$.

T10 Only $K(y,y,z)$ is defined; add $K(a,b,z)=\frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}$.

Remark In the commented-out BFKL section the logarithm is written log $\frac{k^+}{\Lambda}$. The BFKL terms multiply log $\frac{\vee}{k^+}$: $d^3N_{\rm LL}=\mathcal{N}\big[\log\frac{k^+}{\Lambda}\,\text{JIMWLK}\otimes\text{LO}+\log\frac{\vee}{k^+}\,\text{BFKL}\otimes\text{LO}\big]$. This is changed in the corrected copy; it has no effect on the PDF.

5 Reproducing

In nlo_gluon_paper/evol_check/:

```
python3 cmp_two.py        # two-rho rows vs Group rows (J)           -> cmp_two.json
python3 cmp_three.py      # three-rho rows (both forms) vs Group rows -> cmp_three.json
python3 cmp_list.py       # 'All three rho' list, total, blocks      -> cmp_list.json
python3 terms_check.py    # Sec.1 + Sec.2 = Sec.3 = JIMWLK x LO      -> terms_check.json
python3 measures.py       # measures vs integrands
python3 make_corrected.py; python3 make_corrected.py marked   # corrected/
EVOL_DIR=$PWD/corrected python3 cmp_two.py   # also cmp_three.py, cmp_list.py
```